Survey Report: Emerging Trends in Low-Code Development

1. Introduction

Low-code development is a way of creating software with very little coding. It uses visual tools, drag-and-drop features, ready-made parts, forms, and workflows. It helps people build applications faster and with less programming. Today, low-code platforms are also adding AI, automation, cloud services, data analysis, and other modern features.

2. Objectives of the Survey

- To understand what low-code development means.
- To identify the latest trends in low-code development.
- To understand the role of Artificial Intelligence in low-code platforms.
- To study the benefits and problems of low-code development.
- To understand the future of low-code development.

3. Survey Methodology

This survey is based on a study of low-code development, current software development methods, business applications, and new technology trends. The main areas studied are AI, automation, cloud computing, citizen development, security, data analysis, mobile apps, and system integration.

4. Emerging Trends in Low-Code Development

4.1 AI-Powered Low-Code Development

AI is becoming an important part of low-code platforms. It can help create forms, workflows, database structures, and simple business logic. This can save time and reduce manual work.

4.2 Natural Language Development

Users can describe what they want in simple English instead of writing code. For example, a user can ask for a student attendance application, and an AI tool may create the basic application structure.

4.3 Generative AI

Low-code platforms are adding Generative AI features such as chatbots, text creation, document summaries, automatic replies, and smart suggestions. These features make applications more useful.

4.4 Hyperautomation

Hyperautomation means using different technologies together to automate complete business tasks. Low-code tools can work with AI, RPA, workflows, and APIs to reduce manual work.

4.5 More Use in Companies

Many companies are using low-code tools to create employee systems, customer systems, inventory systems, approval systems, dashboards, and other business applications. It helps companies build applications faster.

4.6 Citizen Development

Citizen development allows people who are not professional programmers to create simple applications. For example, employees in HR, finance, or marketing can create applications for their own work.

4.7 Low-Code and Traditional Coding Together

Low-code does not completely replace professional programmers. Simple applications can be built with low-code, while professional developers can handle complex coding, security, integrations, and performance.

4.8 Cloud-Based Low-Code Development

Many low-code platforms work through the cloud. Cloud-based platforms make it easier to access, deploy, update, and scale applications.

4.9 Better Security

As more important applications are built with low-code, security is becoming more important. Platforms now provide features such as user permissions, login controls, data protection, and activity records.

4.10 Data Analysis and Reports

Low-code platforms can create charts, dashboards, reports, and other data views. This helps organizations understand their data and make better decisions.

4.11 Mobile Applications

Low-code tools can also be used to create mobile applications. Users can build forms, notifications, data entry screens, and other mobile features with less coding.

4.12 API and System Integration

Low-code applications can connect with databases, CRM systems, ERP systems, payment systems, cloud services, and AI services. This makes it easier for different systems to work together.

5. Survey Findings

Trend	Main Benefit
AI	Faster application development
Generative AI	Build applications using simple language
Automation	Reduces manual work
Citizen development	Allows non-programmers to build apps
Cloud	Easy access and deployment
Security	Better protection and control
Data analysis	Better reports and decisions
Mobile development	Easy mobile app creation
API integration	Connects different systems

6. Benefits of Low-Code Development

- Faster development – Applications can be built in less time.
- Lower cost – Less coding work can reduce development cost.
- Easy to use – People with basic technical knowledge can build simple applications.
- Better productivity – Developers can spend more time on difficult tasks.
- Quick changes – Applications can be changed quickly when business needs change.

• Less manual work – Workflows can automate repeated tasks.

7. Challenges

• Vendor lock-in – It may be difficult to move an application to another platform.

• Limited customization – Very complex applications may still need normal coding.

• Security risks – Poorly designed applications may create security problems.

• Performance – Some applications may need extra work when handling very large amounts of data.

• Training – Users still need to understand basic application design and security.

• Governance – Companies need rules for creating, testing, and managing applications.

8. Low-Code vs Traditional Development

Feature	Low-Code	Traditional Coding
Development speed	Usually faster	Usually slower
Coding	Less coding	More coding
Customization	Good for many tasks	Very high
Ease of use	Easy for beginners	Needs programming skills
Cost	Often lower	Often higher
Complex applications	May need extra coding	Well suited
Deployment	Usually faster	May take more time

9. Future Scope

In the future, low-code platforms may use more AI to create complete applications from simple instructions. They may also provide automatic testing, error finding, security checking, data analysis, and workflow creation. Low-code, AI, cloud computing, and automation are likely to work more closely together.

10. Conclusion

Low-code development is changing the way applications are built. It helps organizations create software faster with less coding. AI, Generative AI, automation, cloud computing, data analysis, and system integration are making low-code platforms more powerful. Low-code is useful for simple and medium-level business applications, while professional developers are still needed for complex systems. With proper security and rules, low-code development can become an important part of modern software development.

11. References

• Research articles and reports on Low-Code and No-Code Development.

• Academic studies on Low-Code Application Development.

• Reports and documentation on Artificial Intelligence and Generative AI.

• Research on Robotic Process Automation and Hyperautomation.

• Studies on Cloud-Based Application Development.